# AP Physics C: App Blueprint

Interactive Learning Platform Architecture & Development Guide

## 1. Executive Summary & Philosophy

This document serves as the foundational blueprint for developing an interactive, self-paced AP Physics C learning application. The objective is to translate the extreme rigor of *Physics by Halliday, Resnick, and Krane (HRK)* and MIT OCW into a highly accessible, scaffolded digital experience. The platform will lower the barrier to entry to university-level physics by systematically replacing passive textbook reading with interactive workflows, logic checks, and micro-assessments.

## 2. Core System Architecture (The Four Engines)

The platform will be built upon four distinct "engines" that handle content delivery, interaction, and assessment.

### Engine 1: Scaffolding & Support

The backend logic designed to prevent student frustration and cognitive overload.

- **The "Hint Cascade" System:** Progressive hints for complex problems (e.g., Hint 1: Conceptual framing → Hint 2: Mathematical setup → Hint 3: First integration step).
- **The "Why It's Wrong" Engine:** Interactive decision trees that catch common mistakes (e.g., forgetting the chain rule) and dynamically route users to micro-refreshers.
- **Calculus "Training Wheels":** 2-minute modular math refreshers seamlessly embedded within physics workflows.

### Engine 2: Interactive Concept Builders

Digital, step-by-step tools that replace traditional textbook reading.

- **FBD & Drag Force Workflows:** Drag-and-drop vector builders preceding differential equation setup.
- **Gauss's Law Architect:** A 3D simulator requiring students to select appropriate Gaussian surfaces and set integral limits.
- **Energy Landscape Maps:** Visual tools where students input $U(x)$ to derive force graphs and identify equilibria.
- **Biot-Savart Builders:** Interactive setups forcing directionality checks (cross-products) before calculating magnetic fields.

### Engine 3: Media Curation Hub

Targeted, bite-sized delivery of rigorous university-level material.

- **Textbook Excerpt Digitizer:** Clean, focused text selections from HRK 5th Edition, highlighting core derivations.
- **MIT OCW Micro-Clips:** Embedded timestamps of specific conceptual leaps or demonstrations from MIT 8.01/8.02, skipping lengthy blackboard proofs.

### Engine 4: AP Testing & Strategy Engine (April-May)

A specialized mode unlocked near exam time focused entirely on the College Board format.

- **"Speed-Run" MCQ Trainers:** Pattern recognition drills relying on dimensional analysis and limiting cases.
- **FRQ Rubric Simulators:** Interactive grading exercises where the student acts as the College Board grader on sample answers.
- **Endurance Mode:** Timed, strict simulation exams with scaffolding and hints completely disabled.

## 3. Month-by-Month Development Roadmap

This outlines the curriculum sequencing and the interactive modules to be built each month.

## Mechanics Build Phase (September – December)

### September: Kinematics & Dynamics

**Curation:** HRK Chapters 2-6.
**Interactive Build:** "Drag Force Differential" — FBD creation to variable separation.

### October: Energy & Momentum

**Curation:** HRK Chapters 7-10.
**Interactive Build:** "Energy Landscape" — deriving $F = -dU/dx$ and 2D collision logic trees.

### November: Rotational Mechanics

**Curation:** HRK Chapters 11-13.
**Interactive Build:** "Rolling Without Slipping" decision matrices and cross-product trainers.

### December: Gravitation & Oscillations

**Curation:** HRK Chapters 16-17.
**Interactive Build:** Physical pendulum derivation scaffolding (identifying center of mass and moment of inertia).

## E&M Build Phase (January – March)

### January: Electrostatics

**Curation:** HRK Chapters 25-28.
**Interactive Build:** "Gauss's Law Architect" and charge density integral simulators.

### February: Circuits & Magnetostatics

**Curation:** HRK Chapters 29, 31-34.
**Interactive Build:** RC circuit differential equations; Biot-Savart vector setup tools.

**March: Electromagnetism**

**Curation:** HRK Chapters 35, 36, 38.
**Interactive Build:** Faraday's / Lenz's Law simulators for predicting induced current direction.

## 4. The Build & Test Cycle (Weekly Workflow)

A structured 7-day iteration loop to ensure high-quality, low-friction modules.

**Days 1-2: Resource Extraction**
Scan/digitize HRK text and problem sets. Locate exact MIT OCW timestamps.

**Days 3-4: Interactive Architecture**
Build the digital workflow logic. Sequence excerpts, embed clips, and write interactive checkpoints.

**Days 5-6: Stress Testing**
Act as the student. Audit the workflow for steep logical jumps or undocumented mathematical assumptions.

**Day 7: Refinement**
Based on the stress test, lower the barrier to entry by adding intermediary steps or calculus "training wheels."

## 5. Foundational Resources

The app content heavily relies on acquiring and structuring the following primary texts:

- **Physics by Halliday, Resnick, and Krane (5th Edition) - Volume 1:** Covers Mechanics, Wave Motion, and Thermodynamics.
- **Physics by Halliday, Resnick, and Krane (5th Edition) - Volume 2:** Covers Electricity, Magnetism, and Optics.
- *Sourcing:* Digital copies can be referenced via academic libraries, The Internet Archive, or purchased as official Wiley E-Texts for digitization purposes.